



Low-Cost High Safety Microcontroller: AVR® SD

ASIL C and SIL 2 Compliant

Achieving high safety integrity levels (SIL) for safety-critical functions in automotive and industrial applications is complex and requires expertise and rigorous processes. Functional safety engineers face pressure to reduce design costs to stay competitive. Selecting the right microcontroller is crucial to meet international functional safety standards and optimize costs without compromising safety. To address this, we announced the AVR® SD family, compliant with ISO 26262 (ASIL C) and IEC 61508 (SIL 2). It is ideal for safety functions like monitoring temperature, current, or voltage spikes and simplifies designs by offloading safety-critical functions from a central processor. With built-in hardware safety features (dual cores in lockstep, ECC on memory, a dedicated error controller), it detects faults within 1 millisecond and includes a complete development ecosystem with compliant software, safety documentation, and a certified compiler.

Key Features

- **Dual-Core Lockstep CPU:** Ensures high diagnostic coverage of faults by running two identical processing cores in parallel and catching any discrepancies almost instantaneously (1 clock cycle)
- **Error Correcting Code (ECC) on All Memories:** Detects and corrects errors, maintaining data integrity
- **Dedicated Error Controller Module:** Continuously monitors for errors and takes corrective actions, maintaining system integrity and safety
- **Voltage and Clock Monitors:** Ensures the microcontroller operates within safe voltage and clock parameters, preventing system failures due to power or timing issues
- **Fault Detection Time Intervals:** Suitable for applications with high safety demands and fault detection time interval targets down to 1 millisecond

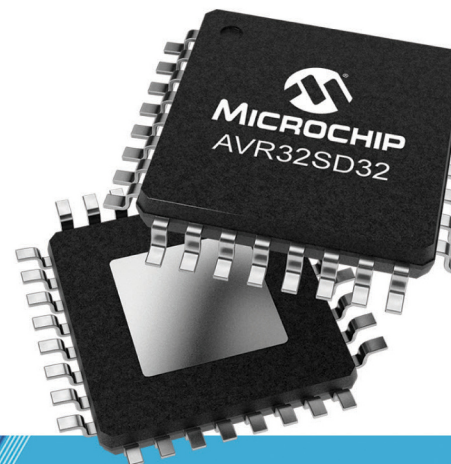
Join the Future of Safety-Critical Design With AVR32SD32.

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